

ECON 351 - Diversification & Variance Problems

Screenshots from homework, sample exam, mock midterm, lecture, and practice tests

Sources checked:

- Homework 1 Ch.4: No diversification/variance problems
- Homework 2 Ch.9: No diversification/variance problems
- Sample Questions Ch.4: No diversification/variance problems
- Sample Questions Ch.9: Question 22 - Feng/Nan portfolios (included)
- Mock Midterm 1: Question 29 - John/Marc variance (included)
- Econ351 Lecture Ch.9: Variance, correlation, diversification slides (included)
- Practice Test 1 Q24: Same as Sample Q22
- Practice Test 2 Q24: Same as Mock Q29
- Practice Test 3 Q24: Same as Sample Q22

6 problem sets in this PDF.

Variance, Correlation & Diversification - Formula Sheet

Expected value: $E[x] = p_1 \cdot x_1 + p_2 \cdot x_2 + \dots + p_n \cdot x_n$

Variance: $\text{Var}(x) = \sum (p_i \cdot (x_i - E[x])^2)$

Higher variance = more risk / more volatile outcomes

Correlation: do variables move together? (positive / negative / zero)

Diversification: combine assets that are NOT fully positively correlated

Negatively correlated portfolio -> lower variance than either asset alone

Positively correlated portfolio -> higher variance (less diversification benefit)

Problem 1: D-1

Sample Questions Ch.9 - Question 22 | Practice Tests 1 & 3 Q24

Feng: agriculture (A & B). Nan: airline/oil (C & D).

Question 22:

Feng has all her wealth invested in Company A and in Company B. These are agricultural companies, and their value can go up or down depending on the amount of rainfall this season. There is a 50% chance that

will rain a lot, in which case the value of her investment in Company A will go up to \$100 while her investment in Company B will go up to \$90. There is a 50% chance of a drought, in which case the value of her investment in Company A will go down to \$80, while her investment in Company B will go down to \$70.

Nan has her wealth invested in Company C and in Company D. Company C is an airline company, while Company D is an oil company. Their value can go up or down depending on the international price of oil. There is a 50% chance that the price of oil will go up, in which case her investment in Company D will go up to \$100 while her investment in Company C will go down to \$70. There is a 50% chance that the price of oil will go down, in which case the value of her investment in Company D will go down to \$80, while her investment in Company C will go up to \$90.

- Are the values of Company A and Company B positively or negatively correlated?
- Are the values of Company C and Company D positively or negatively correlated?
- Who has the riskier investment portfolio, Feng or Nan?

Problem 2: D-2

Mock Midterm 1 - Question 29 | Practice Test 2 Q24

John: half A + half B (positively correlated). Marc: half C + half D (negatively correlated).

Question 29:

Assets A and B have positively correlated returns, while assets C and D have negative correlated returns. John is an investor who has chosen to invest half of his money on asset A and half on asset B. Marc is an investor who has chosen to invest half of his money on asset C and half on asset D. Therefore, which

of the following options is most likely to be true when we compare John and Marc?

- (A) John's investment has a lower variance and John is more risk averse than Marc
- (B) John's investment has a higher variance and John is more risk averse than Marc
- (C) John's investment has a lower variance and John is less risk averse than Marc
- (D) John's investment has a higher variance and John is less risk averse than Marc
- (E) Both investment portfolios have the same variance and the investors are risk neutral.

Problem 3: D-3

Econ351 Lecture Ch.9 - Variance (formula)

Lecture slide: variance definition and formula.

STATISTICS - VARIANCE

We use the **variance** as a measure of **riskiness**:

$$\text{Var}(x) = p_1(x_1 - E(x))^2 + p_2(x_2 - E(x))^2 + \dots + p_n(x_n - E(x))^2$$

$$\text{Var}(x) = \sum_{i=1}^n (p_i(x_i - E(x))^2) = \sigma^2$$

- ▶ A constant variable has a variance of zero
- ▶ A large variance indicates more uncertainty:

Problem 4: D-4

Econ351 Lecture Ch.9 - Variance (numeric examples)

Compare Investment 1 (low variance) vs Investment 2 (high variance).

STATISTICS - VARIANCE

We use the **variance** as a measure of **riskiness**:

$$\text{Var}(x) = p_1(x_1 - E(x))^2 + p_2(x_2 - E(x))^2 + \dots + p_n(x_n - E(x))^2$$

$$\text{Var}(x) = \sum_{i=1}^n (p_i(x_i - E(x))^2) = \sigma^2$$

- ▶ A constant variable has a variance of zero
- ▶ A large variance indicates more uncertainty:

Investment 1: 50% chance of \$310, or 50% chance of \$290

$$\text{Var}(\text{Investment1}) = 0.5(310 - 300)^2 + 0.5(290 - 300)^2 = 100$$

Investment 2: 50% chance of \$600, or 50% chance of \$0

$$\text{Var}(\text{Investment2}) = 0.5(600 - 300)^2 + 0.5(0 - 300)^2 = 90000$$

Problem 5: D-5

Econ351 Lecture Ch.9 - Correlation

Correlation measures whether variables move together.

STATISTICS - CORRELATION

Another important concept is **correlation** between variables:

- ▶ Loosely speaking, variables that move together

Example 1: Daily temperature and sales of ice cream. In days with higher temperatures, more ice cream is sold

Problem 6: D-6

Econ351 Lecture Ch.9 - Diversification

Diversified portfolio (Nvidia + Sony) reduces risk vs single stock.

DIVERSIFICATION

Another way to reduce risk is **diversification**:

- ▶ Allocating resources to a variety of activities whose outcomes are not fully correlated

Value of a \$10,000 Investment After One Year		
	AI Boom Continues (50% chance)	AI Bubble Bursts (50% chance)
Nvidia (high AI exposure)	14,000	7,500
Sony (gaming/hardware)	8,500	11,500

Diversifying to hold both an AI-growth stock and a stock hurt by AI's reduces risk

Similarly, holding a diversified portfolio of stocks reduces risk

- ▶ i.e. buy an index fund, not the stock you 'think' is going to do well

Answer Key

D-1:

a) A & B positively correlated. b) C & D negatively correlated. c) Feng has riskier portfolio.

D-1 detail:

Feng: \$190 if rain, \$150 if drought (EV=\$170, high spread). Nan: always \$170 (diversified).

D-2:

(D) John's investment has higher variance and John is less risk averse than Marc.

D-3/D-4:

$\text{Var}(\text{Inv1})=100$; $\text{Var}(\text{Inv2})=90,000$. Same expected value can have very different risk.

D-5:

Positive correlation: variables move together. Correlation does not imply causation.

D-6:

Hold assets with outcomes not fully correlated to reduce portfolio risk.